HALİÇ UNIVERSITY / FACULTY OF ENGINEERING / DEPARTMENT OF COMPUTER ENGINEERING
2024 - 2025 ACADEMIC YEAR - SPRING SEMESTER
CEN401 - NUMERICAL ANALYSIS FINAL EXAM - 22/05/2025

Name & Surname :

Student Number :

Signature :

INSTRUCTIONS:

- The exam will start at **17:00** and will finish at **18:30**. The total time of the exam is **90** minutes.
- This exam consists of six questions. Please explain your answers step by step.
- Write **four** digits after the comma point.

Good luck!

QUESTIONS

Q1. What are interpolation and extrapolation in numerical methods and why do we use them? What is the difference between them? Explain briefly. (5 pts)

Answer:

Q2. Find Lagrange interpolating polynomials that approximate $f(x) = x^3$.

a. The linear interpolation polynomial $P_1(x)$ using the nodes $x_0 = -1$ and $x_1 = 0$. (4 pts)

b. The quadratic interpolation polynomial $P_2(x)$ using the nodes $x_0 = -1$, $x_1 = 0$, and $x_2 = 1$. (6 pts)

c. The cubic interpolation polynomial $P_3(x)$ using the nodes $x_0 = -1$, $x_1 = 0$, $x_2 = 1$, and $x_3 = 2$. (8 pts)

Answer:

TY / FACULTY OF ENGINEERING / DEPARTMENT OF COMPUTER ENGINEERING
EMIC YEAR / SPRING SEMESTER
RICAL ANALYSIS FINAL EXAM - 22/05/2025

Evaluate the integral

$$\int_0^{2.4} \frac{2x}{1+x^2}\,dx$$

using the following methods.

a. Simpson's $\frac{1}{3}$ method. Divide the whole interval into six subintervals. (10 pts)

b. Simpson's $\frac{3}{8}$ method. Divide the whole interval into six subintervals. (10 pts)

c. The exact value of the integral is $I = \ln\left(\frac{169}{25}\right)$. Compare the results and discuss the reasons for the differences. (7 pts)

(Advice/Hint: It might be a good idea to create a table first.)

Answer

Q4:

a):

Given $f(x) = e^x$, we approximate $f'(2.3)$ using finite difference methods.

For $h = 0.1$:

Forward difference:

$$f'(2.3) \approx \frac{f(2.4) - f(2.3)}{0.1} = \frac{e^{2.4} - e^{2.3}}{0.1} = \frac{11.0232 - 9.9742}{0.1} = 10.49$$

Backward difference:

$$f'(2.3) \approx \frac{f(2.3) - f(2.2)}{0.1} = \frac{9.9742 - 9.0250}{0.1} = 9.492$$

Central difference:

$$f'(2.3) \approx \frac{f(2.4) - f(2.2)}{0.2} = \frac{11.0232 - 9.0250}{0.2} = 9.991$$

For $h = 0.01$:

Forward difference:

$$f'(2.3) \approx \frac{f(2.31) - f(2.3)}{0.01} = \frac{10.0741 - 9.9742}{0.01} = 9.99$$

Backward difference:

$$f'(2.3) \approx \frac{f(2.3) - f(2.29)}{0.01} = \frac{9.9742 - 9.8747}{0.01} = 9.95$$

Central difference:

$$f'(2.3) \approx \frac{f(2.31) - f(2.29)}{0.02} = \frac{10.0741 - 9.8747}{0.02} = 9.97$$

b):

The exact value of $f'(2.3)$ is $e^{2.3} = 9.9742$. Among all approximations, the central difference with $h = 0.1$ gave the closest result as 9.991. When h is too small, round-off errors may affect accuracy despite smaller truncation errors. Central difference generally yields better results due to higher-order accuracy, and the best result here was indeed related to both the method and appropriate choice of h.